

ULTRON 2.7 Phase 3 Execution Report

Dataset quality and safety regression layer

Project	ULTRON 2.7
Phase	Phase 3: Dataset Quality And Safety Evaluation
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 3 adds dataset quality and safety evaluation tooling. ULTRON can now audit the supplied synthetic dataset, identify ambiguous duplicate labels, flag underspecified commands, extract safety regression cases, and evaluate planner/policy behavior against those cases.

Phase 3 Scope

- Analyze the full 2,500-example dataset.
- Detect duplicate utterances with conflicting labels.
- Flag underspecified search/open-file examples.
- Extract high-risk and confirmation-gated safety cases.
- Evaluate policy behavior on the safety regression set.

What Was Implemented

Area	File(s)	Result
Dataset quality module	src/ultron27/dataset_quality.py	Reusable analyzer and safety-case extraction.
Quality script	scripts/analyze_dataset_quality.py	Writes JSON report and safety regression JSONL.
Safety evaluator	scripts/evaluate_safety_regression.py	Checks expected tools and policy actions.
Generated artifacts	docs/phase3_dataset_quality_report.json, data/regression/safety_cases.jsonl	Concrete audit output.
Tests	tests/test_pipeline.py	Expanded to 17 regression tests.

Audit Metrics

Metric	Value	Meaning
Total examples	2500	Full supplied synthetic dataset size.
Unique utterances	2319	Normalized utterances after duplicate grouping.
Duplicate groups	101	Utterances appearing more than once.
Conflicting duplicate groups	101	Duplicate groups with different labels or arguments.
Underspecified examples	879	Examples whose labels contain slots not explicit in the text.
Safety examples	362	High-risk, confirmation-gated, or unsupported examples.

Verification Results

Check	Command	Result
Unit tests	<code>python -m unittest discover -s tests</code>	17 tests passed.
Dataset evaluation	<code>python scripts/evaluate_dataset.py --split test</code>	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Quality audit	<code>python scripts/analyze_dataset_quality.py</code>	Report and safety cases generated.
Safety regression	<code>python scripts/evaluate_safety_regression.py</code>	362 cases, tool accuracy 1.0, policy action accuracy 1.0.

Recommended Next Step

- Deduplicate conflicting synthetic rows.
- Convert underspecified file-search rows into clarification examples.
- Add real usage logs with corrected transcripts.
- Add Hinglish, Hindi, and STT-error variants.
- Keep safety cases as a permanent regression gate.